

M-4 in DLD lectures 2024

DATE: / /

PAGE NO.:

Module-2 (Functional Completeness)

In boolean algebra we have seen many operations. Using these operations, we can write bool. exp. of any boolean function.

But, do we really need all of them to write exp. of any boolean function?

No, not all required,

→ Set of boolean operation which is sufficient to write expression of every boolean function is called "functionally complete" set of boolean expression.

Ex: {NOR}, {AND, OR, NOT}, etc.

→ Set of logic gates which are enough to implement any digital circuit is called "functionally complete" set of gates.

→ Similar def. for functionally comp. set of set theory operations & functionally complete set of prop. logic operations.

⇒ Proof that {AND, OR, NOT} is func. comp. in boolean alg.
As we can write SOP / POS for any function
i.e. we can express any func. using {AND, OR, NOT},
So it's functionally complete.

⇒ Proof that {AND, NOT} / {NOT, OR} is func. complete.

As using demorgan's law $AND \rightarrow OR, NOT$ & $OR \rightarrow AND, NOT$, So, {AND, NOT} or {OR, NOT} is also functionally complete.

Superset of FC set is obviously functionally complete.

DATE: / /

PAGE NO.:

Ex: Remove OR.

$$A+B \Rightarrow \overline{\overline{A+B}} \Rightarrow \overline{\overline{A} \cdot \overline{B}} \quad \text{(No OR)}$$

Remove AND

$$\overline{A} \cdot \overline{B} \Rightarrow \overline{\overline{\overline{A} \cdot \overline{B}}} \Rightarrow \overline{\overline{A} + \overline{\overline{B}}} \Rightarrow \overline{\overline{A} + B} \quad \text{(No AND)}$$

So {AND, NOT}, {OR, NOT} both are functionally complete.

be2:

Any fun.

Truth

Table

SOP/POS

Remove AND

using demorgan's law

Only OR, NOT

Remove OR

Only AND, NOT

Ques: Eliminate addition from f:

$$f = x(\overline{y} + \overline{z}) + \overline{x}y$$

$$f = x(\overline{\overline{y} + \overline{z}}) + \overline{x}y$$

$$f = x(\overline{yz}) + \overline{x}y$$

$$f = \overline{x} \overline{yz} + \overline{x}y$$

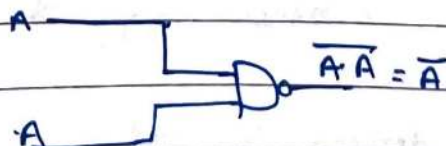
$$f = \overline{x} \overline{yz} \cdot \overline{\overline{y}} + \overline{x}y$$

$$f = \overline{x} \overline{yz} \cdot y + \overline{x}y$$

⇒ Prove that NAND Gate is an universal gate.

Proof: To prove that, any bool. fun. can be implemented using only NAND gates, we will show that AND, OR, NOT operation can be performed using only these gates.

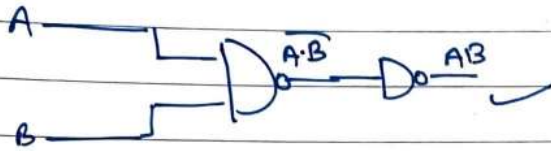
a). NOT Gate.



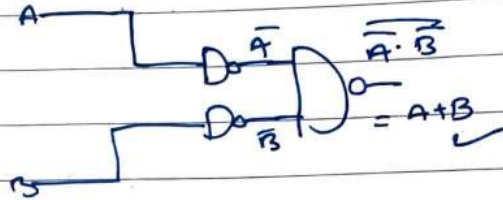
If some FC Set is determined by some Set, then

DATE: / /
PAGE NO.:

B) AND Gate.



C) OR Gate



Hence, Proved.

Universal gates are those gates which can imp. any Boolean function w/o need to use any other gate type.

Ex: NAND Gate, NOR Gate.

We can use only universal gates to make any digital circuit.

Ques: Express $\overline{x+y}$ using only NAND operations.

$$\Rightarrow \overline{x+y} \Rightarrow (\overline{x+y}) \Rightarrow (\overline{x+y}) \Rightarrow (\overline{x+y})$$

$$\Rightarrow ((\overline{x+y}) \uparrow (\overline{x+y})) \uparrow ((\overline{x+y}) \uparrow (\overline{x+y})) \Rightarrow \overline{x+y}$$

Similarly we can prove that NOR is also a universal Gate.

$$\begin{aligned} \text{But } x+y &= (\overline{x+y}) \downarrow (\overline{x+y}) \\ \overline{x} &= \overline{x} \downarrow \overline{x} \end{aligned} \quad \left\{ \begin{array}{l} \text{NAND is derived from NOR} \end{array} \right.$$

DATE: / /

PAGE NO.:

To check if a set of f.c. always minimise operations of set and then try to create {NOT, AND} or {NOT, OR} or any other f.c. set from that.

We also have a formal algorithm for this f.c.

Checking: (Saw later)

⇒ Prove that $\{\rightarrow, \neg\}$ is func. complete.

$$\left. \begin{array}{l} a+b = \neg a \rightarrow b \\ \neg a = \neg a \end{array} \right\} \begin{array}{l} (\neg, \neg) \xrightarrow[\text{from}]{\text{derived}} (\neg, \neg) \\ \quad \quad \quad \hookrightarrow \text{Already f.c.} \quad \quad \hookrightarrow \text{So f.c.} \end{array}$$

⇒ Prove that $\{\rightarrow, \vee\}$ / $\{\rightarrow, \wedge\}$ is not f.c.

→ By formal theorem told later.

⇒ Prove that $\{\rightarrow, 0\}$ is f.c.

$$\neg a = a \rightarrow 0$$

$$a+b = \neg a \rightarrow b = (a \rightarrow 0) \rightarrow b$$

So {OR, NOT} is derived from $\{\rightarrow, 0\}$. So f.c.

* Properties of boolean functions:

a) 0-preserving boolean function: f is 0 preserving iff $f(0, 0, \dots, 0) = 0$.

Ex: $a+b$, ab , $a \oplus b$, a , 0 ↪ constant function

$a \uparrow b$, $a \downarrow b$, $a \odot b$, $\bar{a}$ are not 0 preserving.

$\in T_1$

b) 1-preserving boolean functions: f is 1 preserving iff
 $f(1, 1, \dots, 1) = 1$

i.e. if all ip are 1 then o/p is 1.

Ex: $a+b, ab, a \rightarrow b, a \odot b, a, 1$ are 1 preserving

$a \uparrow b, a \downarrow b, a \oplus b, \bar{a}, 0$ are not 1 preserving.

NOTE $\leftrightarrow$ If we know a fun is 0 preserving, we can't say anything about 1 preserving property or vice-versa.

 $\in S$

c) Self Dual Function: f is self dual if $f = f^d$.

Ex: $a, \bar{a}$ are self dual.

$a+b, ab, 0, 1, a \odot b, a \oplus b, a \uparrow b$ are not self dual func

 $\in L$

d) Linear Function: f is linear if one of the following two statements holds:

a) For each 1-value of f , the no. of 1's in corresponding ip is odd & for every 0-value of f , the no. of 1's in corresponding ip is even.

OR

b) For each 1-value of f , the no. of 1's in corr. ip is even & for every 0-value of f , the no. of 1's in corresponding ip is odd.

(F shouldn't have dummy variables)

If one of the above stmts hold, f is linear.

DATE: / /

PAGE NO.:

Ex: $f = a$

a	f
0	1
1	0

(even 1's)

(odd 1's)

} $\Rightarrow$ Linear.

$f = ab$

a	b	f
0	0	0
0	1	0
1	0	0

(1 1 1)

Not linear bcz. for 0-values

#1's are sometimes odd,

sometimes even. They should

be odd always (here).

 $a \oplus b$
 $a \rightarrow b$
 $a \rightarrow b$
Similarly $a \oplus b$ is also not linear.

$f = a \oplus b$

a	b	f
0	0	0
0	1	1
1	0	1
1	1	0

Linear.

1-value = #1 are odd

0-value : #1 are even.

Similarly $a \odot b$ is also linear.NOTEBefore checking linearity of function to minimise f.
bcz. dummy var. can change answer.For Ex: $f(a, b) = a + ab$

a	b	a+ab
0	0	0
0	1	0
1	0	1
1	1	1

Not linear.

DATE: / /

PAGE NO.:

On minimising

$$f(a,b) = a + ab = a$$

a	a
0	0
1	1

} Linear NowSo, do NOT forget to remove the dummy variable (b).Linear Boolean function $\Leftrightarrow$ Alternating function $\Leftrightarrow$ Counting function

Ques: $f(a,b,c) = \bar{a}b + a\bar{b} + abc$

Check linearity of f . $\Rightarrow$ Minimise f .

$$\bar{a}b + a\bar{b}$$

$$= a \oplus b$$

 $\hookrightarrow$ Linear ✓NOTE Constant functions are not Linear. ✓EN

e) Monotonic Boolean Functions: for function f , if inc. the i/p comb. by changing any single i/p from 0 to 1, then f value should not decrease. (val to 0 not allowed)

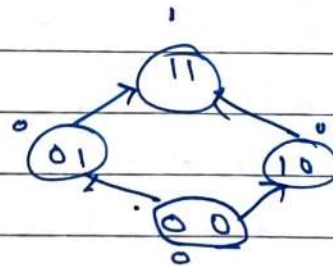
Ex:

a	b	c	f	f'
0	0	1	0	1
0	1	1	0	1
1	0	1	1	0

Best way to check for monotonic function is by making lattice.

Ex: check if $a \oplus b$ is monotonic

a	b	$a \oplus b$
0	0	0
0	1	1
1	0	1
1	1	0



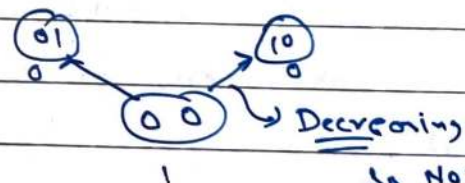
Value should not dec. on this path

So, monotonic

$a \oplus b$ is also monotonic.

Check $a \rightarrow b$ is monotonic or not

a	b	$a \rightarrow b$
0	0	1
0	1	1
1	0	0
1	1	1



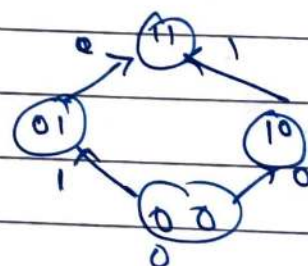
Decreasing
So NOT monotonic

$a \rightarrow b$, $a \oplus b$

$00b$, $a \oplus b$ are also not monotonic.

Check $f(a, b) = b$ is monotonic or not

a	b	b
0	0	0
0	1	1
1	0	0
1	1	1



Monotonic

Lattice for 3-var can also be created.

DATE: / /

PAGE NO.:

NOTE

Constant functions 0 & 1 are monotonic as f value can never be decreased.

★★

Post's Functional Completeness Theorem: A set of boolean functionally complete

if and only if this system doesn't belong to any of entirely

T_0, T_1, L, M, S .

i.e. X is set of func.

$X \not\subseteq T_0$ and $X \not\subseteq T_1$, i.e. $X \not\subseteq L$ & $X \not\subseteq M$ & $X \not\subseteq S$

where T_0 is set of 0 preserving function

T_1 " " " " 1 " "

L " " " " Linear " "

M " " " " Monotonic " "

S " " " " Self-Dual " "

→ to be functionally complete.

i.e. X should have: ~~at least one function~~

- a) at least one function that don't preserve zero, and,
- b) " " " " " " one, and,
- c) " " " " is not linear, and,
- d) " " " " " " monotonic, and,
- e) " " " " " " self-dual.

So, if all ^{not} f are 0 preserving, OR
 " " " " 1 " " OR
 " " " " Linear, OR
 " " " " Monotonic OR
 " " " " Self dual, then
 X is not f.c.

DATE: / /

PAGE NO.:

Ex:1 $X = \{x \leftrightarrow y, xy, 0\}$

Not 0 preserving $\emptyset$
 Not 1 " $\{0\}$
 Not Linear $\{$
 Not Monotonic $\{$
 Not Self Dual $\{x \leftrightarrow y, xy, 0\}$

As all fun. are 0 preserving it is not F.C.Ex:2 $\{x \downarrow y, xy, 1\}$ (Directly, Superset of $xy, 1$ so F.C.) $x \downarrow y, 1, xy$ is not zero preserving xy is not 1 preserving $1, xy$ is not linear $1, xy$ is not monotonic $xy, 1$ is not Self dualEx:3 $\{0, 1, \rightarrow, \leftrightarrow\}$ $\Rightarrow$ All are 1 preserving. So NOT F.C.Ex:4 Show that $\{1, A\bar{B}\}$ is F.C. $\rightarrow$ $1 \rightarrow$ Not 0 preserving, Not self dual $A\bar{B} \rightarrow$ Not 1 preserving, Not monotonic, Not Linear, Not self dual
 $\hookrightarrow 1, 0 \text{ is } 1$
 $\hookrightarrow 1, 1 \text{ is } 0$

So, by Emil Post's theorem,

It is functionally Comp.

Method 2.Create NOT.

$$\overline{A \cdot B} = \overline{A} + \overline{B} \quad \text{NOT can be created}$$

Create AND

$$A \cdot \overline{B} = A \cdot B \quad \text{AND can be created}$$

So F.C.Ques.Ans:

f(a,b,c) given by T.T. below by F.C.

a	b	c
1	1	1
1	1	0
1	0	1
1	0	0
0	1	1
0	1	0
0	0	1
0	0	0

f(a,b,c)

0
0
1
0
1
0
0
1

f*

0
1
1
0
1
0
1
1

→ Comp. & Reverse

Q. Show that f obey's req. of post's completeness theorem.

$\Rightarrow f(1,1,1) = 0$ ✓ Not 1 preserving
 $f(0,0,0) = 1$ ✓ Not 0 preserving
 $f(1,1,0) = f(1,0,0) = 0$ ✓ Not linear (minimized table)
 (as odd 1 & odd 0)

$$f = \overline{a}\overline{b}\overline{c} + \overline{a}b\overline{c} + a\overline{b}c$$

No dummy

Check using K-map.

DATE: / /

PAGE NO.:

$$\begin{aligned} f(0,0,1) &= 1 \\ f(1,1,1) &= 0 \end{aligned} \quad \left. \vphantom{\begin{aligned} f(0,0,1) &= 1 \\ f(1,1,1) &= 0 \end{aligned}} \right\} \text{Not Monotonic}$$

from f^d row $\hookrightarrow$ Not Self Dual.

$$\text{as } f \neq f^d$$

or

$$f(1,1,0) \neq f(0,0,1)$$

So, obeys all req. of Post's f.c. theorem.

b). Show that f is f.c. by deriving any already know f.c. set.

$\hookrightarrow$ Using k-map

$$f(a,b,c) = \bar{a}bc + a\bar{b}c + a\bar{b}\bar{c}$$

$$f(a,a,a) = \bar{a}$$

$\hookrightarrow$ NOT created

$$f(a,a,1) = \bar{a} + c$$

$\hookrightarrow$ NOR created

And {NOT, NOR} is f.c.

So, Hence Proved

NOTE $\longleftrightarrow$ There is NO official/std. term "Partially Functional Comp."

DATE: / /

PAGE NO.:

Imp. Note

In boolean function 0, 1 are not by default available to us so $x \rightarrow y$ is not f.c. y

$\rightarrow$ is bool. function & set $\{1, 0\}$ is f.c.

In digital circuits, 0, 1 are by default available to us. So, gate " $\rightarrow$ " is functionally complete as 0 & 1 are available to us.

So, in Exam take care of, they are asking about f.c. of boolean function or logical gates

^{gate min, adder}
Digital Ckt / Gates $\rightarrow$ 0 & 1 available

Boolean function $\rightarrow$ 0, 1 Not available unless given.

GO CLASSES

These notes are the property of GOClasses. Please do not share them without their consent.

For any questions or issues regarding my handwritten notes, feel free to reach out via email at karan757527@gmail.com or DM me on telegram at @karan757527 (or by clicking [here](#)).

I would love to know if you liked my notes. Your feedback and appreciation are invaluable, so don't hesitate to share your thoughts. Click [here](#) to connect with me on LinkedIn or you can find me as @agrawalkaran.

Wishing you success on your journey ahead!

- Karan Agrawal